

Refreshing ... Uncensored ...
Groundbreaking

The Dr. Wang's original lecture:

Tutorial of Design Compiler

Outline

- Introduction
- Setting Up the Tutorial
- Graphical Interface
- The Alarm Clock Design
- Setting Design Environment
- Setting Design Constraints
- Overview of Optimization Phases
- Analysis of Report

DC Tutorial - 2

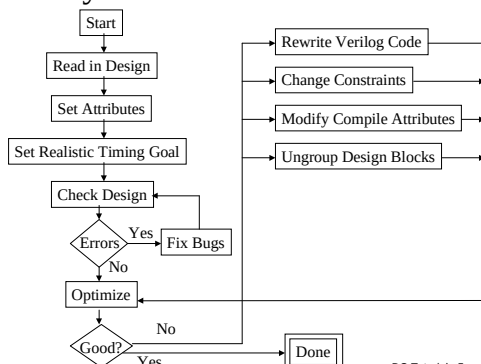
Introduction

Introduction

- The Synthesis Process
- Design Compiler Products
- Synthesis Programs and Tools
- Design Styles
- Input and Output Formats
- User Interfaces
- Script Files

DC Tutorial - 4

The Synthesis Process



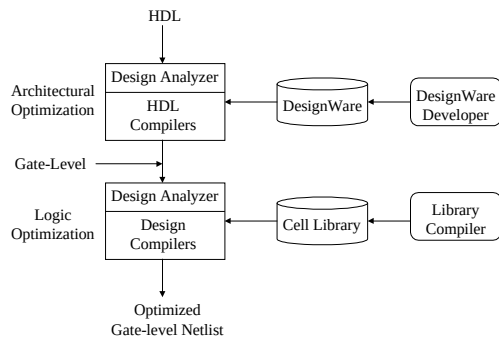
DC Tutorial - 5

The DC Products

- DC Professional
 - No multi-frequency clocking, latch-based time borrowing, pipeline re-timing, critical path re-synthesis, in-place optimization, and incremental editing
- DC Expert
 - Include features for maximizing performance
- FPGA Compiler
 - Targets only FPGA technology

DC Tutorial - 6

Synthesis Tools



DC Tutorial - 7

Architectural Optimization

- Arithmetic Optimization
- Timing and Area-Based Resource Sharing
- Sub-expression Removal
- Constraint-Driven Resource Selection
- Inference of Synthetic Part (DesignWare)
- For more information
 - HDL Compiler for Verilog Reference Manual

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DesignWare

- Provide a library of high-level design components
 - Adders, Multiplier, etc.
- The HDL compiler will select the proper components for you based on your timing and area goals
- See Documentation Collection (open collection)
 - Synopsys DesignWare 1997.01

DC Tutorial - 9

DesignWare Developer

- Create DesignWare Libraries

DC Tutorial - 10

DC Products

- Optimize your design at the gate level
- Using selected cell libraries

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Cell Library

- Library of basic cells used by DC
 - AND, OR, XOR, etc.
- For FPGA compiler, it may contain more complex cells
 - Xilinx CLBs, IOBs, etc.

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Library Compiler

- Yes, you can create your own cell libraries

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Design Styles

- Hierarchical or Flatten
- Combinational or Sequential

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Input Formats

- VHDL
- Verilog
- PLA & EDIF 2.00
- Xilinx XNF

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Output Formats

- Synopsys binary format (.db files)
- VHDL
- Verilog
- EDIF 2.00
- Equation, LSI Logic, Mentor Graphics, PLA, state table, Tegas formats
- Xilinx XNF format

DC Tutorial - 16

User Interfaces

- shell> dc_shell
 - unix-like command shell
 - dc_shell> **quit**
 - dc_shell> **cd my_dir**
 - dc_shell> **alias vv write -f verilog**
 - dc_shell> **pwd**
 - dc_shell> **history n**
 - dc_shell> **list -command**
 - dc_shell> **man <command>**
 - dc_shell> **sh "lpr <filename>"**
- shell> design_analyzer
 - graphical interface

DC Tutorial - 17

Scripts

- A set of command can be put together into a file called "script"
- Then, you don't need to re-type some the commands again and again when using the dc_shell
- Scripts for this tutorial will be provided for your reference
- You can run them when you are home without the X-window capability

DC Tutorial - 18

Locate Documentation

- shell> **design_analyzer &**
- select **Help --> On-Line Documentation**
- Ignore the square window with "Titles"
 - select **Cancel** to close it
 - focus on the one with "File, Edit, View ..."
- select **File --> Open Collection**
- select **Synopsys Synthesis Tools 1997.01** and then click **OK**
- select **Documents Formatted for Printing** and then click **Open**
- In the "File, Edit, View ..." window, now you can select a list of on-line documents

DC Tutorial - 19

Setting Up the Tutorial

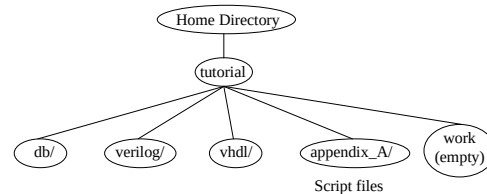
Setting Up the Tutorial

- Creating the directories
- Setting paths and aliases
- Creating a start-up file
- Running tutorial with scripts

DC Tutorial - 21

Creating The Directories

- **>cp -r /baby/synopsys/doc/syn/tutorial .**



DC Tutorial - 22

Path

- **%> source /usr/local/bin/setup.synopsys**
- Or you can put it in .cshrc file
 - **%> source .cshrc**
 - **%> rehash**
- You can take a look of the setup file
 - **%> more /usr/local/bin/setup.synopsys**
- It basically setup the right environmental variables for you

DC Tutorial - 23

.synopsys_dc.setup file

- Creating a .synopsys_dc.setup file can overwrite system default settings
- **%> cp ~/tutorial/.synopsys_dc.setup ~/.synopsys_dc.setup**
- **%> vi ~/.synopsys_dc.setup**
 - company = "Motorola Somerset";
 - designer = "CEO";
 - view_background = "while";

DC Tutorial - 24

More about setup file

- `%> more .synopsys_dc.setup`
 - `search_path = { } + search_path`
 - `link_library ... target_library ... symbol_library ...`
 - `define_design_lib ...`
- `search_path = {a directory} + search_path`
 - if you cp tutorial into a *directory* other than home
- `link_library`: location of subdesigins referenced by the design
- `target_library`: identify technology libraries
- `symbol_library`: identify symbols library for generating/viewing schematics
- `define_design_lib`: identify a temporary place to store intermediate files created by the analyzer

DC Tutorial - 25

Scripts

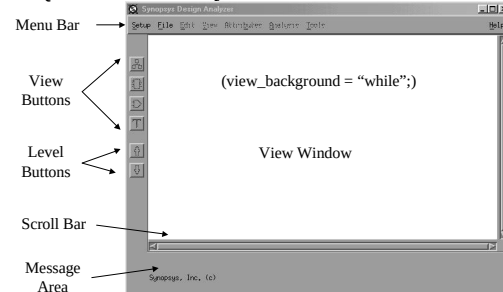
- No X-Window, No Problem
- Find script files in `~/tutorial/appendix_A/`.
- See *Design Analyzer Reference Manual* for more detail

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Graphical Interface

Graphical Interface

- Start: `%> design_analyzer &`
- Quit: Select File --> Quit



DC Tutorial - 28

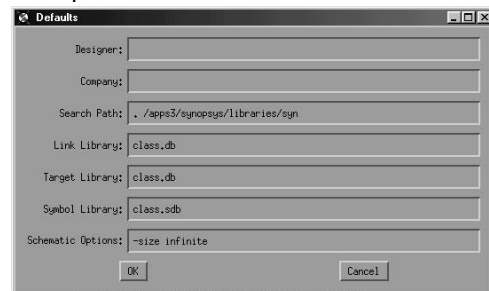
Mouse Buttons

- Left Button
 - Select design and design objects
- Mid Button
 - Add or remove objects from a group of objects already selected
- Right Button
 - Bring up the pop-up menu

DC Tutorial - 29

Check Default Setup

- Setup --> Defaults ...



DC Tutorial - 30

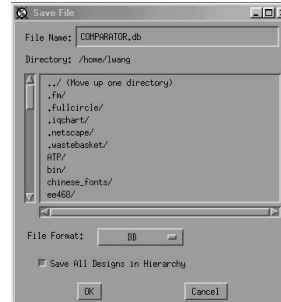
Read in a Design

- File -->
 - analyze & elaborate
 - read
- analyze :
 - read in VHDL/Verilog; check for syntax and synthesize logic; store as intermediate formats
 - Use to read each sub-design + top level design
- elaborate :
 - create the design from intermediate formats; determine the correct bus size; Use for top level design + sub-design with parameters passing in
- read : read design formats other than HDL (db, PLA, etc.)

DC Tutorial - 31

Save a Design

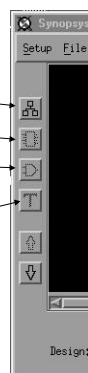
- File --> Save or Save As ...
 - Once a design is selected



DC Tutorial - 32

A Design Has 4 Views

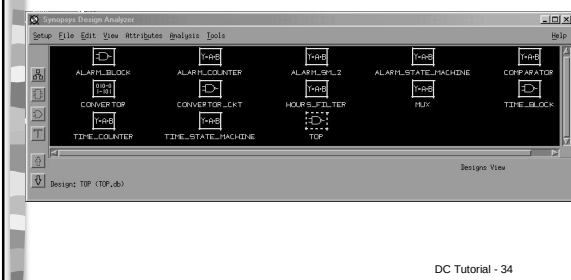
- Design View
- Symbol View
- Schematic View
- Hierarchy View
- T View (No Use)



DC Tutorial - 33

Design View

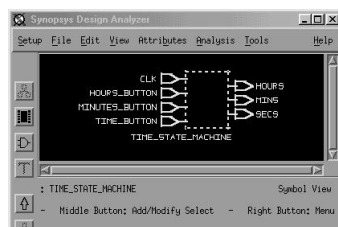
- After "read" in all 13 verilog files in the tutorial directory
 - you first enter the Design View



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Symbol View

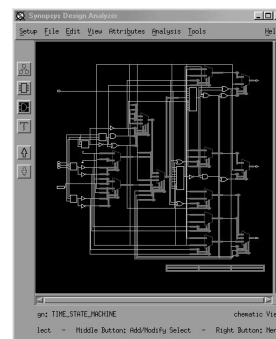
- Select TIME_STATE_MACHINE and double-click on it -> you enter the symbol view of the design



DC Tutorial - 35

Schematic View

- Click on the "schematic view" button on the left hand side



DC Tutorial - 36

Hierarchy View

- Click the **up arrow** (left hand side) to go back to design view; double-click on **TOP**; Select **View --> Change View --> Hierarchy**
 - TOP contains 6 modules



DC Tutorial - 37

Design View Icons

- Netlist:** read in as a netlist and optimized to gates
- Equation:** In VHDL, Verilog, or equation format that is partially or completely behavioral
- PLA:** Specified in PLA format
- State Table:** Specified in state table format



Netlist



Equation



PLA

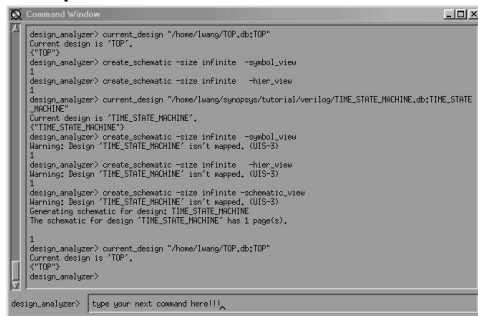


State Table

DC Tutorial - 38

Command Window

- Setup --> Command Window ...**



DC Tutorial - 39

dc_shell Commands

- For more information,
 - see *Design Compiler Reference Manual: Fundamentals*

DC Tutorial - 40

Design Attributes

- Attributes are values you set to control the optimization process
 - Select **Attributes** from the menu
- The Attributes menu provide access to
 - Set input and output delays
 - Set drive strengths
 - set loads
 - Characterize subdesigns
 - Select operating conditions
 - Choose a wire load model
 - Create or modify a clock

DC Tutorial - 41

Operating Environment Sub-menu

- Set design properties that describes the internal conditions of a design and the design's interaction with its surrounding
 - drive strength on ports
 - the time that signals arrive on ports
 - load driven by output ports

DC Tutorial - 42

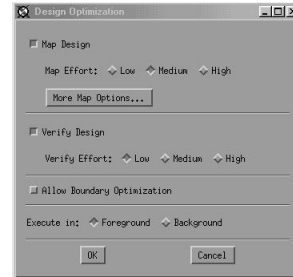
Optimization Constraints

- Set the goal for design optimization
 - largest delay allowed
 - greatest area allowed
- Two set-constraint windows
 - Design Constraints window
 - Goals for area and power
 - Design rules implied by technology library
 - Test-related constraints (testability)
 - Timing Constraints window
 - Timing constraints

DC Tutorial - 43

Design Optimization

- Select Tools --> Design Optimization
 - See *Design Compiler Reference Manual: Optimization and Timing Analysis* for more detail



DC Tutorial - 44

Locating Problems

- Before and after optimization, use Schematic View and Check Design to locate problems
 - Generate schematic view
 - Select Analysis --> Check Design
 - Jump to a design object
 - Click on an error or warning message in the Design error window
 - Click on the show button



DC Tutorial - 45

Generate Report

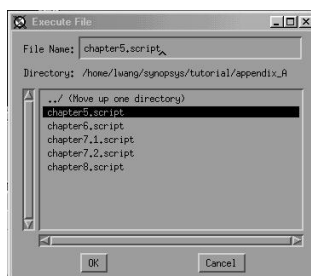
- Analysis --> Report ...



DC Tutorial - 46

Run a Script File

- Setup --> Execute Script
 - check out ~/tutorial/appendix_A/*.script
 - dc_shell> include <script_filename>



DC Tutorial - 47

The Alarm Clock Design